

ENTERPRISE INFRASTRUCTURE PLAN

for ABC Startup Solutions Inc.

A Proposal for Initial IT Hardware, Network, and Security Deployment

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ITEP 414 — System Administration and Maintenance

Bachelor of Science in Information Technology

Week 2 Portfolio Project — Enterprise Infrastructure Planning for a Startup Company

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Document Version 1.0

August 2026

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1. Executive Summary

ABC Startup Solutions Inc. is a newly formed software development company preparing to move into its first office space in Calamba City, Laguna. The company will begin operations with twenty employees spread across four departments and, at present, owns no computers, no server, no internal network, and no formal security policy. This document lays out the infrastructure that needs to be in place before day one of operations, along with the reasoning behind each recommendation.

The plan covers hardware and software procurement, the design of a segmented office network, an on-premises server paired with select cloud services, a layered approach to security, and a backup and recovery strategy the company can actually sustain with a small IT team. It also includes a realistic eight-week rollout schedule, a first-year budget estimate, and a short discussion of the risks the company should watch for once the network goes live.

The overall goal is straightforward: give ABC Startup Solutions an infrastructure that is dependable enough to run daily operations, secure enough to protect client and company data, and flexible enough to grow as headcount increases over the next two to three years. Every recommendation in this document was chosen with that balance of reliability, security, and cost in mind, rather than defaulting to the most expensive or most complex option available.

2. Introduction and Objectives

Most infrastructure problems that companies deal with later on, slow networks, data loss, security incidents, unmanaged software licenses, can be traced back to decisions made or skipped during the very first setup. ABC Startup Solutions has the advantage of starting from a blank slate, which means there is an opportunity to build the network and support structure correctly from the start instead of patching an ad hoc setup after the fact.

This document was written from the perspective of a Junior System Administrator tasked with producing the company's first IT Infrastructure Plan for review by management before any purchase orders are issued. It is meant to be read by non-technical decision makers as well as by whoever eventually manages the network day to day, so explanations are kept practical and tied back to how the company actually operates rather than written as abstract technical theory.

Objectives of This Plan

This plan sets out to achieve the following:

- Translate the operational needs of four departments, IT, HR, Finance, and Sales, into a concrete list of hardware, software, and network equipment.
- Design a network that keeps departmental traffic reasonably separated while still allowing shared resources such as printers and file storage.
- Establish security controls that are appropriate for a twenty-person company, not an enterprise with a dedicated security team, without leaving obvious gaps.
- Define a backup and disaster recovery approach that the company can realistically test and maintain.
- Provide a rollout timeline and budget that management can use to plan procurement and cash flow.
- Leave the design room to grow so that the network does not need to be rebuilt the next time the company hires.

3. Business and Technical Requirements

3.1 Company Profile

ABC Startup Solutions Inc. is a software development company that builds web and mobile applications and provides ongoing IT consulting for small and medium enterprises in the CALABARZON region. The company was formed to give local SMEs access to the kind of custom software support that is normally reserved for larger clients, at a price and turnaround time that smaller businesses can actually work with.

Vision: To be the software development partner that CALABARZON-based small and medium enterprises turn to first, known for dependable delivery and for building long-term relationships rather than one-off projects.

Mission: ABC Startup Solutions delivers well-tested, maintainable software on realistic timelines, mentors its junior staff into senior technical roles, and treats every client engagement as if the company's own reputation depends on it, because it does.

Office Location (fictional): Unit 4B, 4th Floor, Camella Business Center, Brgy. Real, Calamba City, Laguna. The unit is approximately 250 square meters on a single floor, laid out as an open workspace with a separate server closet, a small meeting room, and a pantry area.

3.2 Organizational Structure and Employee Distribution

The company is led by a Managing Director, under whom four department leads report directly. Each department lead is responsible for their team's day-to-day output and works with the IT function on any equipment, access, or software requests specific to their group. The table below shows the confirmed headcount used as the basis for every hardware and network calculation in this document.

Department	Employees	Primary Function
Information Technology	5	Software development, DevOps, internal IT support, and network administration
Human Resources	4	Recruitment, employee records, payroll coordination, and office administration
Finance	5	Billing, accounts payable and receivable, budgeting, and financial reporting
Sales	6	Client acquisition, account management, and field visits to prospective clients
Total	20	

3.3 Technical Requirements Derived From the Scenario

Because the company currently owns no equipment of any kind, every requirement below had to be derived directly from how each department works rather than assumed from an existing setup. IT and part of Sales need machines capable of running development tools and virtual machines. HR and Finance handle sensitive personal and financial records and need controlled access and reliable backup more than raw computing power. Sales staff spend a meaningful part of their week outside the office, which pushes their device mix toward laptops rather than desktops. These differences are reflected in the hardware and network inventories in the sections that follow.

4. Current and Proposed Infrastructure

4.1 Current State

The company's current infrastructure is, in practical terms, empty. There are no desktops, laptops, servers, switches, or wireless access points, and the office unit has not yet been fitted with structured cabling. This is actually a favorable starting point for planning purposes, since there is no legacy equipment or existing cabling to work around and every recommendation in this plan can be sized correctly the first time.

4.2 Proposed Infrastructure Overview

The proposed setup follows a fairly conventional small-office design: a fiber internet connection terminates at an ISP-supplied modem, which connects to a combined router and firewall appliance, which in turn feeds a managed core switch. From the core switch, structured CAT6 cabling runs to workstations, the wireless access points, the network printers, and the on-premises server and NAS unit. Each department is placed on its own VLAN so that, for example, a compromised Sales laptop cannot directly reach Finance file shares. The full topology is shown in Section 5.

On the compute side, a single on-premises Ubuntu Server handles file sharing, DNS, DHCP, and internal Git hosting, while day-to-day productivity work happens on a mix of Windows 11 Pro desktops and laptops. This combination keeps licensing costs reasonable while still giving the IT team a Linux environment to build development and testing skills on, which lines up with the software inventory the company already plans to standardize on.

4.3 Enterprise Hardware Inventory

The hardware list below assumes one primary device per employee, with a small pool of spare desktops and a spare laptop kept by IT for onboarding new hires and covering repairs without disrupting anyone's workday. Device type per department was chosen based on how mobile that department's work actually is, desk-bound roles receive desktops, and roles that travel or move around the office frequently receive laptops.

Asset ID	Hardware	Qty	Department	Purpose
HW-001	Desktop Computer	4	IT	Development workstations for backend, frontend, and QA work
HW-002	Desktop Computer	3	HR	Records management, recruitment tools, and payroll coordination
HW-003	Desktop Computer	4	Finance	Accounting software, spreadsheets, and financial reporting
HW-004	Desktop Computer	2	Sales	Office-based account management and client documentation
HW-005	Desktop Computer	2	IT (Spare Pool)	Onboarding buffer and loaner units during repairs
HW-006	Laptop	1	IT	Lead engineer, on-call support and off-site troubleshooting
HW-007	Laptop	1	HR	HR Manager, off-site interviews and confidential meetings

Asset ID	Hardware	Qty	Department	Purpose
HW-008	Laptop	1	Finance	Finance Manager, bank and client meetings outside the office
HW-009	Laptop	4	Sales	Field sales reps for client visits and presentations
HW-010	Laptop	1	IT (Spare Pool)	Loaner for travel or repair coverage
HW-011	Server (Rack-mount)	1	IT	File, DNS, DHCP, internal Git, and backup services
HW-012	Router / Firewall Appliance	1	IT	Internet gateway, NAT, VPN, and perimeter filtering
HW-013	Managed Switch (24-port)	1	IT	Core switch, VLAN trunking, and inter-VLAN routing
HW-014	Unmanaged Switch (8-port)	1	IT	Secondary switch for the printer and access point cluster
HW-015	Network Printer	2	All Departments	Shared printing, one per office wing
HW-016	UPS (Rack-mount, 1500VA)	1	IT	Power protection for the server and network rack
HW-017	UPS (Standalone, 650VA)	2	Finance, Reception	Safe shutdown protection for critical workstations
HW-018	Wireless Access Point	3	All Departments	Full-floor Wi-Fi coverage for staff and guest SSIDs
HW-019	NAS Storage Unit (8TB, RAID 5)	1	IT	Shared department drives and local backup target
HW-020	External Backup Drive (4TB)	2	IT	Rotating offsite backup copies
HW-021	Monitor (24-inch)	20	All Departments	One per desktop, plus five for laptop users at fixed desks

The desktop-to-laptop ratio was set deliberately: IT and Sales carry the most laptops because their work regularly moves outside a fixed desk, while HR and Finance stay mostly desktop-based since their work involves sensitive records that are easier to keep secure on machines that never leave the building. The two spare desktops and one spare laptop give IT enough headroom to onboard a handful of new hires or swap out a failed unit without placing an emergency order.

4.4 Enterprise Software Inventory

Software standardization keeps support simple for a two-to-three-person IT team and avoids the licensing sprawl that tends to happen when every department picks its own tools. The list below reflects the baseline software every workstation and server will carry at go-live.

Software	Version	License	Purpose
Windows 11 Pro	24H2	Volume license, per device	Primary desktop OS for all non-server workstations
Ubuntu Server	24.04 LTS	Free and open source	Core server OS for file, DNS, DHCP, and Git services

Software	Version	License	Purpose
Microsoft 365 Apps (Office)	Current channel	Business Standard, per user	Word, Excel, Outlook, and Teams for daily office work
Visual Studio Code	1.9x	Free (MIT license)	Primary code editor for the development team
Git	2.4x	Free and open source	Source control for all software projects
GitHub Desktop	3.x	Free	Simplified Git workflow for junior developers
VirtualBox	7.x	Free (GPL)	Local virtual machines for testing and training
Google Chrome	Current release	Free	Standard web browser across all departments
Microsoft Defender for Business	Current release	Included with Microsoft 365 Business Premium	Centrally managed endpoint protection
AnyDesk	8.x	Business license, per technician	Remote support for staff working off-site
7-Zip	24.x	Free (LGPL)	File compression and archive management

Ubuntu Server was chosen over a Windows Server license for the core infrastructure role for two practical reasons. First, it removes a recurring server licensing cost that a twenty-person company does not need to carry. Second, since the IT team already works in Linux for parts of the development stack, running the company's own file and directory services on Ubuntu Server gives them a low-risk environment to build that expertise on before it matters on a client project.

5. Network Architecture

The network follows a star topology built around a single managed core switch, which is the standard, low-maintenance choice for a single-floor office of this size. Rather than placing every device on one flat network, each department is assigned its own VLAN. This keeps normal department traffic, print jobs, file access, internal chat, contained to that department, and it means a problem on one VLAN, a misconfigured device or a compromised laptop, does not automatically have a direct line to every other department's traffic.

ABC Startup Solutions — Enterprise Network Topology

Physical and logical layout of the proposed office network

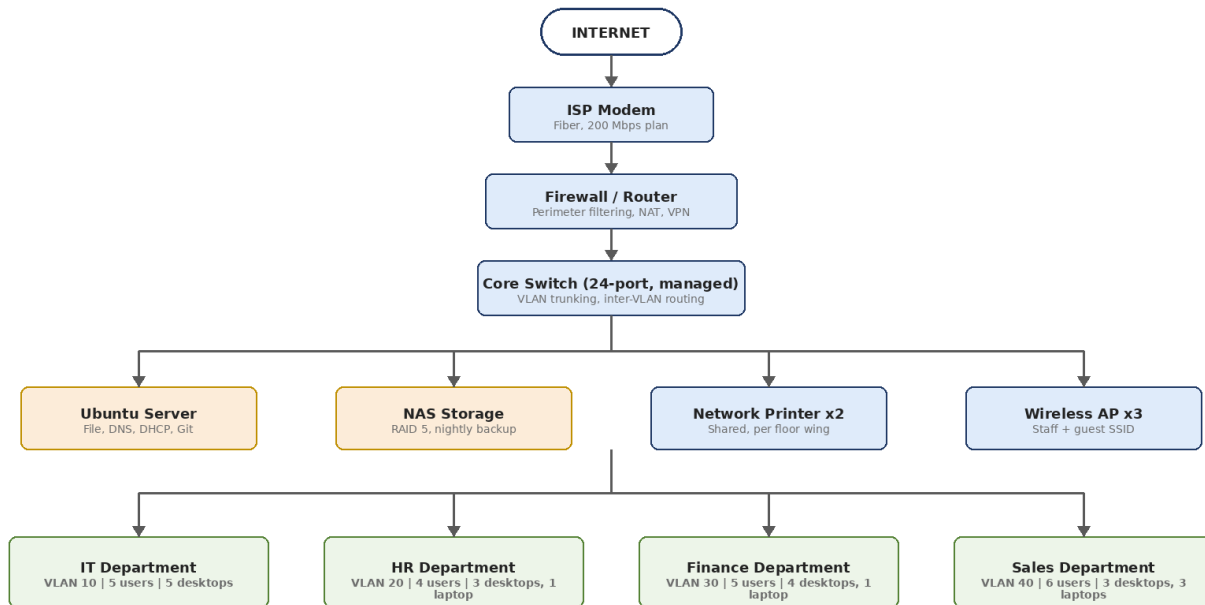


Figure 1. Proposed star topology with VLAN segmentation by department
Prepared by Jaypee Rabie | ABC Startup Solutions Enterprise Infrastructure Plan

Figure 1. Proposed network topology showing the path from the internet connection down to each department VLAN.

5.1 Enterprise Network Inventory

The table below lists the networking equipment required to build the topology shown above, along with realistic quantities for a single-floor, twenty-person office.

Item	Qty	Specification	Purpose
ISP Modem	1	Fiber ONT, gigabit-capable	Terminates the ISP fiber connection into the office
Router	1	Gigabit WAN/LAN, VPN-capable	Internet gateway and inter-VLAN routing
Firewall	1 (integrated)	Stateful inspection, IPS license	Perimeter security, bundled into the router appliance rather than a separate box
Managed Core Switch	1	24-port, Layer 3, VLAN support	Backbone switch for all wired department connections
Secondary Switch	1	8-port, unmanaged	Local switch for the printer and access point cluster
Wireless Access Point	3	Wi-Fi 6, dual SSID (staff / guest)	Full-floor wireless coverage

Item	Qty	Specification	Purpose
Patch Panel	1	24-port, rack-mounted	Termination point for structured cabling runs
CAT6 Cable	2 boxes (305m each)	Solid copper, plenum-rated	Structured cabling for roughly 25 wall drops plus patch cords
RJ45 Connectors	200 pcs	CAT6-rated	Terminating both ends of cable runs, with spares for repairs

The Firewall is listed separately, as required, but in practice it is a licensed feature running on the same router appliance rather than a second physical box. This is a common and cost-effective approach for a network of this size. If the company later needs a dedicated hardware firewall, for example after adding a second office location, the router can be repurposed as an internal routing device at that point.

6. Server and Compute Infrastructure

A single rack-mount server running Ubuntu Server 24.04 LTS will handle the company's core infrastructure services: internal DNS and DHCP, a shared file directory structure organized by department, and a lightweight self-hosted Git remote for projects that should not live on a public GitHub account. Recommended baseline specifications are a six-core server processor, 32 GB of ECC RAM, and a mix of solid-state storage for the operating system and services with larger-capacity drives in a RAID 1 mirror for shared file storage.

Running these services on a single physical box is a reasonable choice at twenty employees. A hypervisor is not recommended at this stage, since it adds a layer of management complexity that a one-to-two person IT team does not need yet. If the company later wants to isolate services, for example running a separate internal test server, VirtualBox is already part of the standard software list and can host that on the same hardware without a full virtualization platform.

Why Ubuntu Server Over a Windows Server License

Beyond the licensing savings mentioned in Section 4, Ubuntu Server's package management and configuration files are straightforward to document and version, which matters for a small company that cannot yet afford dedicated infrastructure-as-code tooling. Long Term Support releases like 24.04 also receive security patches for five years, which keeps the server on a predictable, low-maintenance patch cycle.

7. Storage and Backup

Shared storage sits on the NAS unit listed in the hardware inventory, configured in RAID 5 so that a single failed drive does not take department file shares down with it. Each department gets its own folder with permissions matched to its VLAN and user group, and IT retains a separate administrative share for system images, configuration backups, and installation media.

7.1 Backup Strategy

Backups follow the widely used 3-2-1 approach: three copies of important data, on two different types of media, with one copy stored off-site. In practice, this means the working files live on the server and NAS, a nightly automated backup job copies changed files to one of the two rotating external drives, and that drive is swapped weekly with a second drive kept off-site, initially at a director's residence until the company's client base justifies a proper offsite or cloud archive contract.

Backup Layer	Frequency	Retention	Storage Target
Live working data	Continuous	N/A	Server and NAS (RAID 5)
Local automated backup	Nightly, 11:00 PM	14 days rolling	Onsite external backup drive
Offsite rotation	Weekly, every Friday	8 weeks rolling	Second external drive, stored offsite
Critical records archive	Monthly	12 months	Encrypted cloud storage (see Section 8)

Backups are only useful if they can actually be restored, so a quarterly restore test is built into the maintenance calendar in Section 12. This involves restoring a sample of files from the most recent backup set to a test folder and confirming they open correctly, a step that is easy to skip but is the only way to know a backup strategy is actually working before the day it is needed for real.

8. Cloud and Hybrid Infrastructure

ABC Startup Solutions does not need a full cloud migration at twenty employees, but a purely on-premises setup would leave a few gaps that cloud services fill more cheaply than buying additional hardware. The recommended approach is hybrid: core file and directory services stay on the local server, while a small number of cloud services cover collaboration, source control redundancy, and offsite backup.

Service	Provider	Purpose
Email and collaboration	Microsoft 365 Business Standard	Company email, shared calendars, Teams for internal and client meetings
Source control (client-facing repos)	GitHub	Hosting for client project repositories, separate from the internal Git server
Offsite encrypted backup	Cloud object storage (e.g. Backblaze B2 or equivalent)	Monthly archive copy of critical HR and Finance records
Remote support	AnyDesk	IT access to staff devices working outside the office

This split keeps monthly cloud costs predictable while still giving the company real protection against a worst-case scenario, such as a fire or theft that takes out both the server and the onsite backup drive at the same time. As the company grows, the balance can shift further toward cloud services, but there is no reason to pay for capacity the company does not need in its first year.

9. Security Architecture

Security for a company this size does not need to be complicated to be effective. It needs to be layered, so that no single missed patch or one weak password becomes a company-wide incident. The diagram below outlines the six layers of control recommended for ABC Startup Solutions, moving from the network perimeter down to how activity is monitored day to day. This layered approach reflects widely used defense-in-depth guidance, including recommendations published by the SANS Institute (2022) and the control families described in NIST SP 800-53 (National Institute of Standards and Technology, 2020).

ABC Startup Solutions — Security Architecture (Defense in Depth)

Layered controls from the network perimeter down to data and monitoring



Figure 2. Security controls are layered so that no single failure exposes company data
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Figure 2. Layered security controls from the network perimeter down to monitoring and incident response.

9.1 Key Controls by Layer

- **Perimeter:** the router/firewall appliance filters inbound traffic and blocks unsolicited connections from the internet; a site-to-site or client VPN is used for any remote administrative access.
- **Network segmentation:** VLANs described in Section 5 limit lateral movement between departments and isolate the guest Wi-Fi from the internal network entirely.
- **Endpoints:** Microsoft Defender is deployed on every workstation and the server, with centrally reviewed scan results rather than relying on each user to check their own device.
- **Identity and access:** covered in detail in Section 10, including multi-factor authentication and role-based permissions.
- **Data protection:** laptop drives are encrypted with BitLocker, RAID protects the NAS from single-drive failure, and the backup plan in Section 7 protects against loss or ransomware.

- **Monitoring:** the router, switch, and server logs are checked weekly at minimum, with UPS and connectivity alerts configured to notify IT immediately rather than being discovered after the fact.

9.2 Baseline Policies

A short written acceptable use and security policy will be distributed to all staff during onboarding, covering password requirements, acceptable use of company devices, and the process for reporting a lost device or a suspected phishing email. Documenting these rules, even briefly, gives the company something concrete to point to if a security incident does happen, and it sets expectations before problems occur rather than after.

10. Identity and Access Management

Every employee receives a single named account used across the domain login, Microsoft 365, and internal file shares, rather than shared logins or generic accounts. This keeps an audit trail on who accessed what, which matters most in Finance and HR where records are sensitive, and it means disabling access for a departing employee is a single action instead of a scramble to remember every system they had access to.

10.1 Role-Based Access Control

Access to shared folders and internal systems follows department membership by default. IT staff receive elevated access needed for administration, while HR and Finance shares stay restricted to their respective departments plus the Managing Director. Sales staff have access to shared client documentation but not to HR or Finance folders, which matches how the departments actually work together day to day.

10.2 Multi-Factor Authentication

Multi-factor authentication is required on Microsoft 365 accounts and on any remote access into the company network, including AnyDesk sessions and VPN logins. This single control addresses the most common way small companies get compromised, a reused or guessed password, without requiring specialized security hardware.

10.3 Password Policy

The password policy adopted for ABC Startup Solutions follows current guidance favoring length over frequent forced changes, since research summarized by NIST and CompTIA security materials has shown that overly frequent mandatory resets tend to push users toward weaker, more predictable passwords (CompTIA, 2024).

Requirement	Setting
Minimum length	12 characters
Complexity	Mix of letters, numbers, and at least one symbol
Forced rotation	Every 180 days, or immediately if a compromise is suspected
Account lockout	5 failed attempts, 15-minute lockout
Multi-factor authentication	Required for Microsoft 365, VPN, and AnyDesk
Password reuse	Cannot reuse the last 5 passwords

11. Disaster Recovery and Business Continuity

Disaster recovery planning for a company this size should focus on the failures that are actually likely, a failed hard drive, an extended power outage, or a stolen laptop, rather than building an elaborate plan for scenarios the company is unlikely to face in its first few years. The targets below reflect what is realistic to commit to with a two-to-three person IT team and the infrastructure described in this document.

Scenario	Recovery Time Objective	Recovery Point Objective	Response
Server hardware failure	4 business hours	24 hours	Restore from the most recent local backup onto spare hardware or a replacement drive
Accidental file deletion	1 business hour	24 hours	Restore the specific file or folder from the nightly backup
Extended power outage	Immediate (UPS-covered)	0	UPS provides a graceful shutdown window; server resumes automatically on power restoration
Lost or stolen laptop	Same day	Last sync	Remote wipe via device management, password reset, and encrypted drive limits data exposure
Site-level incident (fire, flood)	5 business days	1 week	Restore from the offsite backup drive and cloud archive onto replacement hardware at a temporary location

These targets are documented here so that expectations are set before an incident happens rather than during one. A short, written incident response checklist, who to call, what to disconnect, where the latest backup is stored, will be kept both digitally and as a printed copy in the server closet, since a purely digital plan is not much help if the incident is the network itself being down.

12. Monitoring, Management, and Support Roles

Day-to-day monitoring for a network of this size does not require an enterprise monitoring platform. The router and managed switch both include basic dashboards for bandwidth and device status, the server logs are reviewed on a set schedule, and the UPS units are configured to send an email alert on any power event. This gives the IT team visibility without introducing another system that itself needs to be maintained.

Task	Frequency	Owner
Review server and firewall logs	Weekly	IT
Confirm backup jobs completed successfully	Daily (automated report)	IT
Apply operating system and security patches	Monthly, or immediately for critical patches	IT
Test backup restore	Quarterly	IT
Review user accounts and access levels	Quarterly	IT with department leads
Review firewall and VLAN rules	Semi-annually	IT

12.1 IT Support Roles and Team Structure

The five-person IT department is expected to cover four functional roles as the company grows, though at launch these responsibilities will overlap across a smaller number of actual staff. Defining the roles clearly now makes it easier to hire correctly as the team expands, and it clarifies who owns what when something breaks.

Role	Responsibilities	Key Skills	Common Tools	Relevant Certifications
Helpdesk Technician	First point of contact for staff issues; account resets, printer and workstation troubleshooting, new-hire equipment setup	Customer service, basic OS troubleshooting, ticket documentation	AnyDesk, ticketing system, Windows 11 admin tools	CompTIA A+
Network Administrator	Maintains routers, switches, VLANs, Wi-Fi, and internet connectivity; manages the firewall ruleset	Networking fundamentals, VLAN design, firewall configuration	Router/switch web console, network monitoring tools	CompTIA Network+, CCNA
Linux System Administrator	Maintains the Ubuntu Server, file shares, DNS/DHCP, internal Git, and backup jobs	Linux command line, shell scripting, service configuration	Ubuntu Server, cron, Git, backup utilities	LPIC-1, CompTIA Linux+
Cloud Administrator	Manages Microsoft 365, GitHub organization settings, and offsite cloud backup	Cloud identity management, SaaS administration, basic cost monitoring	Microsoft 365 admin center, GitHub admin settings, cloud storage console	Microsoft 365 Certified: Fundamentals

In practice, these four roles work as one team rather than four silos. A single support ticket, for example a new hire who cannot access their shared drive, might touch the Helpdesk Technician for the initial request, the Network Administrator to confirm VLAN and Wi-Fi access, and the Linux System Administrator to grant the actual folder permission, with the Cloud Administrator involved only if the new hire also needs a Microsoft 365 license assigned. Keeping the roles distinct on paper, even while the same one or two people cover several of them early on, gives the company a clear hiring roadmap for when the IT team eventually grows past its founding members.

13. Scalability and Future Expansion

The infrastructure described in this document is sized for twenty employees but was deliberately chosen to absorb growth without a redesign. The 24-port core switch has open ports beyond current usage, the router and firewall appliance is rated well above the office's current bandwidth needs, and the VLAN structure can accept a fifth or sixth department VLAN without touching the existing ones.

If headcount grows toward fifty employees over the next two to three years, as the company's business plan projects, the main additions would be a second access point for coverage, additional patch panel capacity if the office expands to a second area on the same floor, and a possible move of email and file storage further into the cloud to reduce the load on the single on-premises server. None of these changes require replacing equipment purchased under this plan, they extend it.

The hybrid cloud approach in Section 8 also gives the company a pressure release valve: rather than buying a larger server up front for growth that has not happened yet, additional cloud storage or a second Microsoft 365 tier can be added incrementally as headcount actually increases, which keeps upfront capital spending closer to what the company needs on day one.

14. Implementation and Rollout Plan

Rolling out the infrastructure in phases, rather than all at once, reduces the chance of a mistake in one area, for example incorrect cabling, delaying every other part of the setup. The eight-week schedule below assumes purchase orders are approved in the first week and equipment lead times of one to two weeks, which is realistic for standard business hardware in the current market.

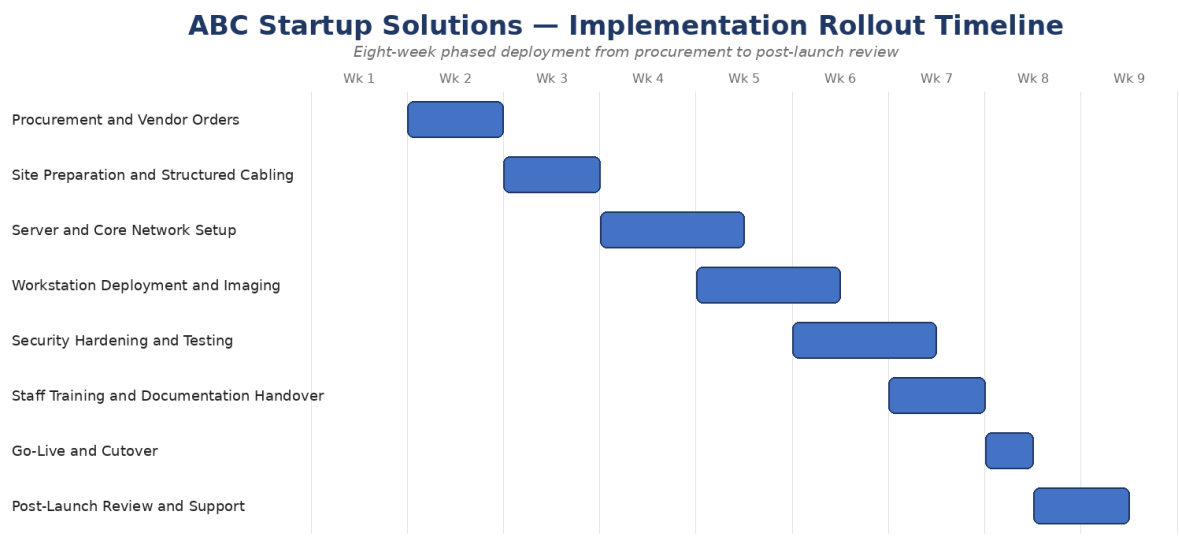


Figure 3. Planned sequence of activities for the initial infrastructure rollout
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Figure 3. Planned eight-week sequence from procurement through post-launch review.

Phase	Weeks	Key Activities
1. Procurement and Vendor Orders	1 – 2	Finalize vendor quotes, issue purchase orders for hardware, networking equipment, and software licenses
2. Site Preparation and Cabling	2 – 3	Install structured cabling, patch panel, and rack in the server closet

Phase	Weeks	Key Activities
3. Server and Core Network Setup	3 – 4.5	Install and configure the router, firewall, switches, access points, and the Ubuntu Server
4. Workstation Deployment and Imaging	4 – 5.5	Unbox, image, and configure desktops and laptops; connect to VLANs and shared drives
5. Security Hardening and Testing	5 – 6.5	Apply firewall rules, deploy endpoint protection, test VLAN isolation, and run the first backup cycle
6. Staff Training and Documentation Handover	6 – 7	Walk each department through their equipment, shared drives, and the acceptable use policy
7. Go-Live and Cutover	7 – 7.5	Full network goes live for daily operations
8. Post-Launch Review and Support	7.5 – 8.5	Monitor closely, address early issues, and formally close out the rollout project

15. Cost and Budget Considerations

The estimate below covers first-year costs, both the one-time capital purchase of hardware and networking equipment and the recurring annual cost of software licenses, internet service, and cloud subscriptions. Figures are approximate, based on current Philippine market pricing for comparable business-grade equipment, and are meant to support budget planning rather than serve as final vendor quotes.

Category	Estimated Cost (PHP)	Notes
Workstations and peripherals	1,150,000	15 desktops, 8 laptops, 20 monitors, keyboards and mice
Server and storage	280,000	Rack server, NAS unit, UPS units, external backup drives
Networking equipment	150,000	Router/firewall, switches, access points, cabling, patch panel
Software licensing (annual)	240,000	Windows 11 Pro, Microsoft 365, Defender for Business, AnyDesk
Internet service (annual)	72,000	Business fiber plan, approximately 200 Mbps
Cloud and backup services (annual)	60,000	Offsite cloud archive storage and related subscriptions
Installation and professional services	120,000	Structured cabling labor, server setup, initial configuration
Contingency (10%)	207,200	Buffer for price fluctuation and unforeseen setup costs
Estimated First-Year Total	2,279,200	

Of this total, roughly 1.58 million pesos is one-time capital expenditure on hardware, networking equipment, and installation, while the remainder is recurring annual cost for software, internet, and cloud services. From year two onward, assuming no major headcount growth, the recurring portion becomes the company's baseline IT operating cost, with capital spending limited to replacement hardware and any expansion covered in Section 13.

16. Risks and Mitigation

No infrastructure plan removes risk entirely, and it would be misleading to present this one as risk-free. The table below lists the risks most relevant to a company of this size and the specific mitigation already built into the design described in earlier sections.

Risk	Likelihood	Impact	Mitigation
Single ISP connection fails	Medium	High	Document a manual failover to a mobile hotspot for critical staff; evaluate a secondary ISP once budget allows
Server hardware failure	Low	High	RAID-protected storage, nightly backups, and a documented restore procedure
Accidental data loss or deletion	Medium	High	Nightly backups with 14-day rolling retention and quarterly restore testing
Malware or phishing attack	Medium	High	Centrally managed endpoint protection, staff awareness training, and MFA on key accounts

Risk	Likelihood	Impact	Mitigation
Weak or reused passwords	Medium	High	Enforced password policy, MFA, and account lockout after failed attempts
Loss of institutional IT knowledge (staff turnover)	Medium	Medium	Written documentation for every system, stored in a shared and access-controlled location
Growth outpacing current capacity	Medium	Medium	Scalable VLAN design and hybrid cloud approach described in Section 13

17. Conclusion

ABC Startup Solutions is starting from nothing, which is exactly the situation where a bit of upfront planning pays off the most. The infrastructure recommended in this document, a segmented office network, an on-premises server paired with a small set of cloud services, layered security controls, and a backup strategy the team can actually maintain, gives the company a foundation that matches its current size without boxing it into that size permanently.

None of the individual pieces in this plan are exotic. What matters is that they were chosen together, with the company's actual departments, budget, and small IT team in mind, rather than assembled from a generic checklist. Followed as written, this plan should get ABC Startup Solutions through its first two to three years of growth without a major infrastructure rebuild, which is the real measure of whether a plan like this one was worth writing.

18. References

The sources below informed the security, identity, and networking recommendations in this document. In-text mentions use an author-year format, for example (NIST, 2020), matching the numbered list below.

- [1] National Institute of Standards and Technology. (2020). Security and Privacy Controls for Information Systems and Organizations (NIST Special Publication 800-53, Revision 5). U.S. Department of Commerce.
- [2] SANS Institute. (2022). Defense in Depth: A Layered Approach to Network Security (Reading Room whitepaper).
- [3] Canonical Ltd. (2024). Ubuntu Server Documentation — 24.04 LTS.
- [4] Cisco Systems, Inc. (2023). Small Business Networking Basics Guide.
- [5] CompTIA. (2024). Security+ and Network+ Certification Exam Objectives.
- [6] Microsoft. (2024). Microsoft 365 and Windows 11 Pro Documentation.

Appendix A — Personal Reflection

Working through this project changed how I think about the System Administrator role. Before this, I mostly thought of the job in terms of fixing things when they broke, resetting a password, replacing a cable, restarting a service. Building this plan from scratch showed me how much of the job actually happens before anything gets plugged in. Deciding how many laptops the Sales team needed, or why the Finance department should stay mostly desktop-based, meant thinking about how people actually work, not just what equipment exists on a spec sheet.

The most challenging part was the network design. It was easy to list the equipment the assignment required, but harder to explain, in a way that would make sense to a non-technical manager, why VLANs mattered for a company that has never had a network incident because it has never had a network. I had to keep asking myself what would actually go wrong if I skipped a control, and use that answer to justify the design instead of just including a feature because it looked thorough. The budget section pushed me the same way: it is one thing to recommend security controls, and another to price them out and see how they compete for the same limited first-year budget as basic workstations.

This project made the case for planning before deployment clearer to me than any lecture could have. Every specification in this document, the switch port count, the VLAN structure, the backup schedule, only makes sense because it was tied back to the twenty-person scenario given in the assignment. If I had jumped straight to buying equipment without that step, I can see exactly how a company would end up with a server that is either too small in six months or wildly oversized for what it actually needed, and a network with no logical structure to expand from later.

Going forward, I think this project will change how I approach even small IT requests. Instead of just fulfilling a request for a new laptop or a new firewall rule, I want to get in the habit of asking what the underlying need actually is, the way this assignment forced me to do for an entire company at once. That habit, translating a business situation into a specific, justified technical decision, feels like the actual core skill this project was built to teach, more than any single tool or diagram in the document itself.